

Hellenic Complex Systems Laboratory

A Software Tool for Estimating Uncertainty of Bayesian Posterior Probability for Disease

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Abstract

Medical diagnosis is central to patient care and healthcare delivery. Established diagnostic practices typically rely on predetermined clinical criteria and numerical thresholds. In contrast, Bayesian inference provides an advanced framework that supports diagnosis via in-depth probabilistic analysis. This study aims to introduce a software tool for quantifying uncertainty in Bayesian diagnosis, which remains insufficiently explored. The presented tool, a freely available specialized software program, utilizes uncertainty propagation techniques to estimate the sampling, measurement, and combined uncertainty of the posterior probability for disease. It features two primary modules and fifteen submodules, all designed to facilitate the estimation and graphical representation of the standard uncertainty of the posterior probability estimates for diseased and nondiseased population samples, incorporating parameters such as the mean and standard deviation of the test measurand, the size of the samples, and the standard measurement uncertainty inherent in screening and diagnostic tests. Our study showcases the practical application of the program by examining the fasting plasma glucose data sourced from the National Health and Nutrition Examination Survey. Parametric distribution models are explored to assess the uncertainty of Bayesian posterior probability for diabetes mellitus, using the oral glucose tolerance test as the reference diagnostic method.

Keywords

Bayesian diagnosis; Bayesian inference; prior probability; posterior probability; likelihood; parametric distribution; probability density function; uncertainty; combined uncertainty; measurement uncertainty; sampling uncertainty; combined uncertainty; confidence intervals; diabetes mellitus; fasting plasma glucose; oral glucose tolerance test

1. Introduction

1.1. Diagnosis in Medicine

Diagnosis in medicine fundamentally involves identifying the unique characteristics of a disease and distinguishing it from other conditions with similar presentations. The term "diagnosis", originating from the Greek word "διάγνωσις" meaning "discernment" (Weiner and Simpson 1989), emphasizes the critical role of distinguishing between healthy and diseased states in individuals. Diagnostic tests are essential in classifying individuals based on their health status. However, the reliance on a singular threshold for diagnosis across a range of data points introduces uncertainty, owing to the overlapping probability distributions of a measurand in healthy and diseased groups (Chatzimichail and Hatjimihail 2023). While traditional diagnostic methods have been broadly effective, they may not fully encompass the diversity of disease manifestations, particularly in varied populations (Choi, Johnson, and Thurmond 2006).

As underlined previously (Chatzimichail and Hatjimihail 2023), Bayesian inference represents a paradigm shift in medical diagnosis, offering a robust framework for integrating various sources of information to make probabilistic assessments. At its core, Bayesian inference relies on Bayes' theorem for updating beliefs in light of new evidence, integrating prior disease probabilities with the distribution of diagnostic measurands to calculate posterior probabilities for disease (Bours 2021; Gelman et al. 2013; van de Schoot et al. 2021; Viana and Ramakrishnan 1992). This approach enables a more comprehensive

probabilistic assessment, the evaluation of the information conveyed by diagnostic measurements, and a personalized patient approach (Choi, Johnson, and Thurmond 2006; Topol 2014).

Historically, the application of Bayesian methods in medicine has undergone significant evolution. Despite facing several challenges and being met with skepticism, these methods have gradually gained acceptance.

1.1.1. Bayes' Theorem in Medical Diagnostics

Bayes' theorem, a fundamental principle in probability theory (Gelman et al. 2013), forms a connection between the direct probability $P(H|E)$ of a hypothesis H given specific data E and the inverse probability $P(E|H)$ of data E given the hypothesis H (Joyce 2021). In medical diagnostics, Bayes' theorem is instrumental in transforming the prior probability for disease into a posterior probability following diagnostic tests (Bours 2021).

1.1.2. Challenges in Applying Bayesian Inference

The application of Bayesian inference in diagnostics, however, faces significant challenges.

Computational Complexity

The computational complexity of Bayesian inference requires considerable resources.

Statistical Distributions in Diagnostics

A significant challenge involves comprehensively understanding the statistical distributions of diagnostic test measurands in both diseased and nondiseased populations (Lehmann and Romano 2008). Calculating posterior probabilities requires probability density functions (PDF) for the measurands in these populations. The normal distribution, often used for simplicity, may not be suitable for measurands with non-normal characteristics like skewness or multimodality. Critical evaluation and potential adoption of alternative distributions are necessary for more accurate Bayesian diagnostic methods (Lehmann and Romano 2008; Box and Cox 1964; D'Agostino and Pearson 1973). *Bayesian Diagnosis*, our previously published software, addresses this challenge (Chatzimichail and Hatjimihail 2023).

Uncertainty of Bayesian Posterior Probabilities

Another significant challenge involves estimating the uncertainty associated with Bayesian posterior probabilities in disease diagnosis. This uncertainty can substantially affect their clinical utility. Despite its critical importance, estimating, evaluating, and mitigating uncertainty in Bayesian diagnostic test interpretation has seldom been addressed in medical literature (Srinivasan, Westover, and Bianchi 2012). To confront this issue, we have developed *Bayesian Diagnostic Uncertainty*, a software tool for estimating uncertainty in Bayesian diagnosis, presented in detail in this study.

Both *Bayesian Diagnostic Uncertainty* and *Bayesian Diagnosis* enhance the applicability of Bayesian methods in medical diagnostics.

1.1.3. Quantifying Uncertainty in Diagnostics

Uncertainty can be quantified and is often expressed probabilistically (Ayyub and Klir 2006).

Combined Uncertainty

In the context of Bayesian posterior probability for disease, we consider two main components of combined uncertainty:

Measurement Uncertainty

This reflects the inherent variability in measurement processes and is defined as a parameter characterizing the dispersion of values that could reasonably be attributed to the measurand (Joint Committee for Guides in Metrology 2011). While crucial for laboratory quality assurance, the impact of measurement uncertainty on clinical decision-making and outcomes is often underexplored and rarely quantified (Kallner et al. 2012; Smith et al. 2019). Emerging research focuses on its effects on misclassification (Ceriotti et al. 2017) and on diagnostic accuracy measures (Chatzimichail and Hatjimihail 2021).

Sampling Uncertainty

The variability in sampling contributes to the uncertainty of posterior probability for disease (Rostron, Fearn, and Ramsey 2020), and it is essential in evaluating diagnostic methods.

2. Methods

2.1. Computational Methods

2.1.1. Bayes' Theorem

Bayes' theorem calculates the posterior probability $P(D|T)$ of a disease D given a test result $T = x$ and a parameter vector θ , as follows:

$$P(D|T) = \frac{f_D(x|\theta)r}{f_D(x|\theta)r + f_{\bar{D}}(x|\theta)(1-r)}$$

Here r denotes the prior probability for disease, $f_D(x|\theta)$ the PDF in disease presence, while $f_{\bar{D}}(x; \theta)$ denotes the PDF in its absence (refer to Appendix A.1 for details).

2.1.2. Parametric Distributions

Parametric statistics operate under the assumption that data from a population can be accurately represented by a probability distribution with a fixed set of parameters (Geisser and Johnson 2006). The program supports the following parametric distributions:

1. Normal distribution
2. Lognormal distribution
3. Gamma distribution.

2.1.3. Uncertainty Quantification

Uncertainty of input parameters can manifest as standard uncertainty $u(x)$, the standard deviation of x , and expanded uncertainty $U(x)$, a range around x encompassing x with a probability p (Kallner et al. 2012).

Measurement Uncertainty

Measurement uncertainty is computed following guidelines in the "Guide to the expression of uncertainty in measurement" (GUM) (Joint Committee for Guides in Metrology 2011) and "Expression of measurement uncertainty in laboratory medicine" (Kallner et al. 2012). Bias is considered a component of this uncertainty (White 2008).

The relationship between the standard measurement uncertainty $u(x)$ to the value of the measurand x , is generally expressed as:

$$u_m(x) = \sqrt{b_0^2 + b_1^2 x^2}$$

where b_0 is a constant and b_1 is a proportionality constant.

If needed, it is approximated linearly as:

$$u_m(x) \approx b_0 + b_1 x$$

The general approach to estimating the coefficients of the above equations is delineated in Appendix A5 of "Quantifying Uncertainty in Analytical Measurement" (Ellison and Williams 2012).

Sampling Uncertainties of Means and Standard Deviations

If m_p and s_p are the mean and standard deviation of a measurand in a population sample of size n_p , then the standard sampling uncertainties of m_p and s_p are estimated as:

$$u_s(m_p) \approx \frac{s_p}{\sqrt{n_p}}$$

$$u_s(s_p) \approx \frac{s_p}{\sqrt{2(n_p - 1)}}$$

using the central limit theorem and the chi-square distribution (Agresti, Franklin, and Klingenberg 2023; Miller and Miller 2018; Aitchison and Brown 1957).

Sampling Uncertainty of Prior Probability for Disease

If n_D and $n_{\bar{D}}$ are the respective numbers of diseased and nondiseased in a population sample, then the standard uncertainty of the prior probability for disease $r = \frac{n_D}{n_D + n_{\bar{D}}}$ is estimated as:

$$u_s(r) \approx \sqrt{\frac{(2 + n_{\bar{D}})(2 + n_D)}{(4 + n_{\bar{D}} + n_D)^3}}$$

using the Agresti–Coull adjustment of the Wald interval (Agresti and Coull 1998).

Combined Uncertainty of Posterior Probability for Disease

The standard combined uncertainty $u_c(x)$ of posterior probability for disease is computed via uncertainty propagation rules, employing a first-order Taylor series approximation (B. M. Wilson and Smith 2013) (refer to Supplementary File II).

When there are I components of uncertainty, with standard uncertainties $u_i(x)$, then:

$$u_c(x) = \sqrt{\sum_{i=1}^l u_i(x)^2}$$

Evidently, $0 \leq u_c(x) < 1.0$.

2.1.4. Expanded Uncertainty of Posterior Probability for Disease

When there are l components of uncertainty, with standard uncertainties $u_i(x)$ and v_i degrees of freedom, then the effective degrees of freedom v_{eff} of the combined uncertainty $u_c(x)$ are obtained from the Welch–Satterthwaite formula (Welch 1947; Satterthwaite 1946):

$$v_{eff}(x) \approx \frac{u_c(x)^4}{\sum_{i=1}^l \frac{u_i(x)^4}{v_i}}$$

If v_{min} the minimum of $v_1, v_2, \dots, v_l$, then:

$$v_{min} \leq v_{eff}(x) \leq \sum_{i=1}^l v_i$$

If $F_v(z)$ is the Student's t -distribution cumulative distribution function with v degrees of freedom and $u_c(x)$ is the standard combined uncertainty of posterior probability for disease, its expanded combined uncertainty $U_c(x)$ at a confidence level p is:

$$U_c(x) \approx \left(F_v^{-1}\left(\frac{1-p}{2}\right) u_c(x), F_v^{-1}\left(\frac{1+p}{2}\right) u_c(x) \right)$$

The confidence interval at the same confidence level p is approximated as:

$$CI_p(x) \approx \left(x + F_v^{-1}\left(\frac{1-p}{2}\right) u_c(x), x + F_v^{-1}\left(\frac{1+p}{2}\right) u_c(x) \right)$$

The confidence intervals of the posterior probability for disease were truncated to the $[0,1]$ range.

2.2. The Software

2.2.1. Program Overview

To facilitate the estimation of the uncertainty of Bayesian posterior probability for disease, the software program *Bayesian Diagnostic Uncertainty* was developed in Wolfram Language, using Wolfram Mathematica® Ver. 13.3 (Wolfram Research, Inc., Champaign, IL, USA). *Bayesian Diagnostic Uncertainty* was designed to estimate and plot the standard sampling, measurement, and combined uncertainty and the confidence intervals of the Bayesian posterior probability for disease of a screening or diagnostic test (See Figure 1).

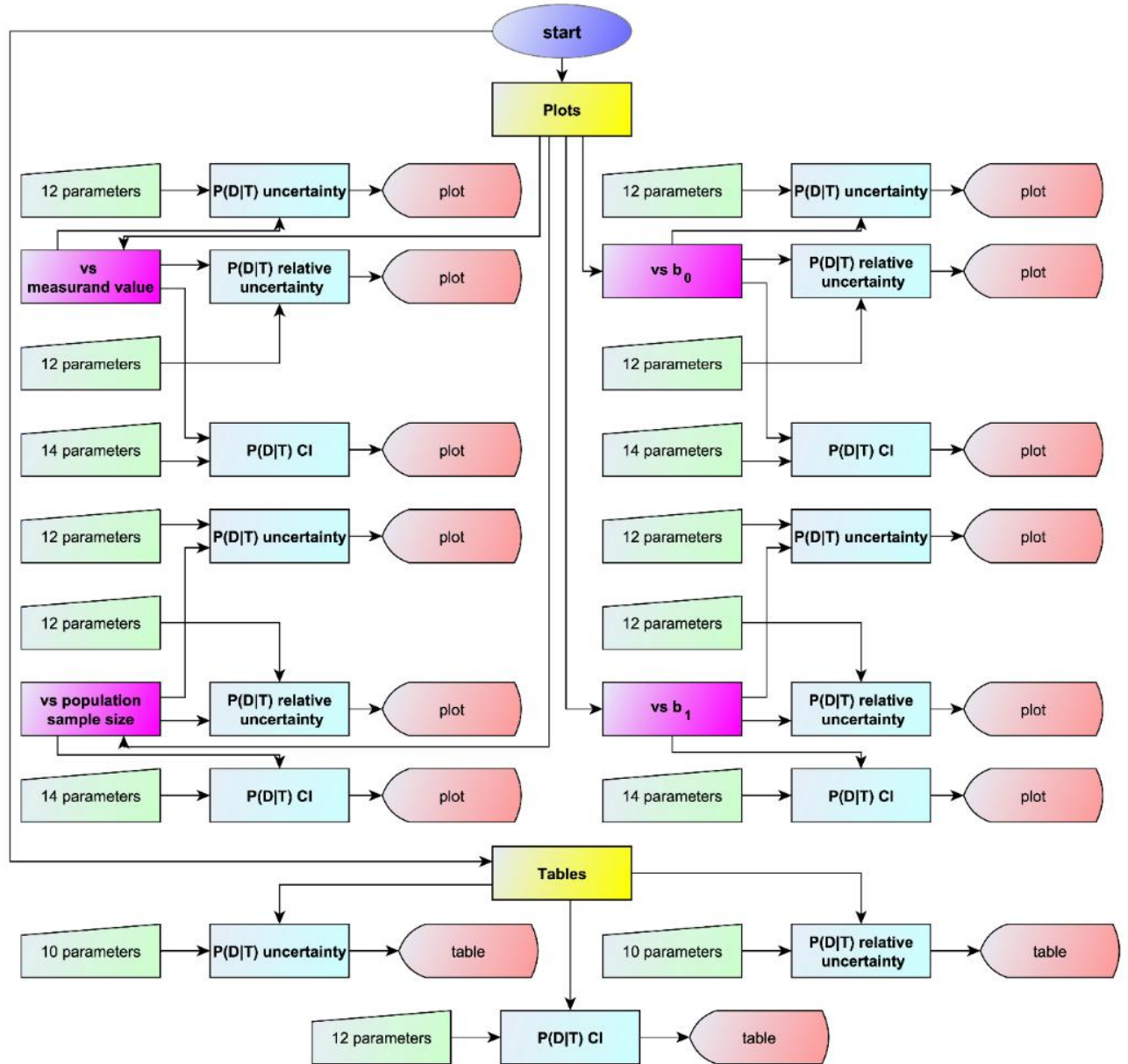


Figure 1. A simplified flowchart of the interface of the *Bayesian Diagnostic Uncertainty* program with the number of input parameters and output types for each submodule.

This interactive program is freely available as a Wolfram Language notebook (.nb) (Supplementary file BayesianUncertainty.nb). It can be run on Wolfram Player® or Wolfram Mathematica® (see Appendix A.2). Due to the complexity of the calculations required, it is computationally intensive.

2.2.2. Input Parameters

The program allows for the definition of three parametric distributions of measurement for diseased and nondiseased populations.

Distribution Selection

The user selects the type of distribution of each population from a predefined list:

1. Normal distribution
2. Lognormal distribution
3. Gamma distribution.

Definition of Statistical Parameters

For each population, the user defines its size n , the mean μ , and the standard deviation σ of the measurand.

Measurement Uncertainty

The user selects a linear or nonlinear equation of the measurement uncertainty versus the value x of the measurand and defines the constant contribution b_0 to the standard measurement uncertainty, the proportionality constant b_1 , and the number of quality control samples analyzed for its estimation.

2.2.3. Output Specifications

Visualizations

The program generates a series of plots designed to elucidate various uncertainty measures and statistics:

1. *Uncertainty of posterior probability for disease*: Plots are generated to show the standard sampling, measurement, and combined uncertainty of the posterior probability for disease.
2. *Relative uncertainty of posterior probability for disease*: Plots are generated to show the relative standard sampling, measurement, and combined uncertainty of the posterior probability for disease.
3. *Confidence intervals of posterior probability for disease*: Plots are generated to show the confidence intervals of the posterior probability for disease for a user defined confidence level.

Tables

For each combination of parametric distributions of the diseased and nondiseased populations, the program tabulates for a user-defined measurand value:

1. The standard sampling, measurement, and combined uncertainty of the posterior probability for disease.
2. The relative standard sampling, measurement, and combined uncertainty of the posterior probability for disease.
3. The confidence intervals of the posterior probability for disease for a user-defined confidence level.

By providing this comprehensive set of input parameters and output specifications (see Figure 2), the program offers a robust platform for exploring the uncertainty in Bayesian disease diagnosis using parametric distributions of medical diagnostic measurands.

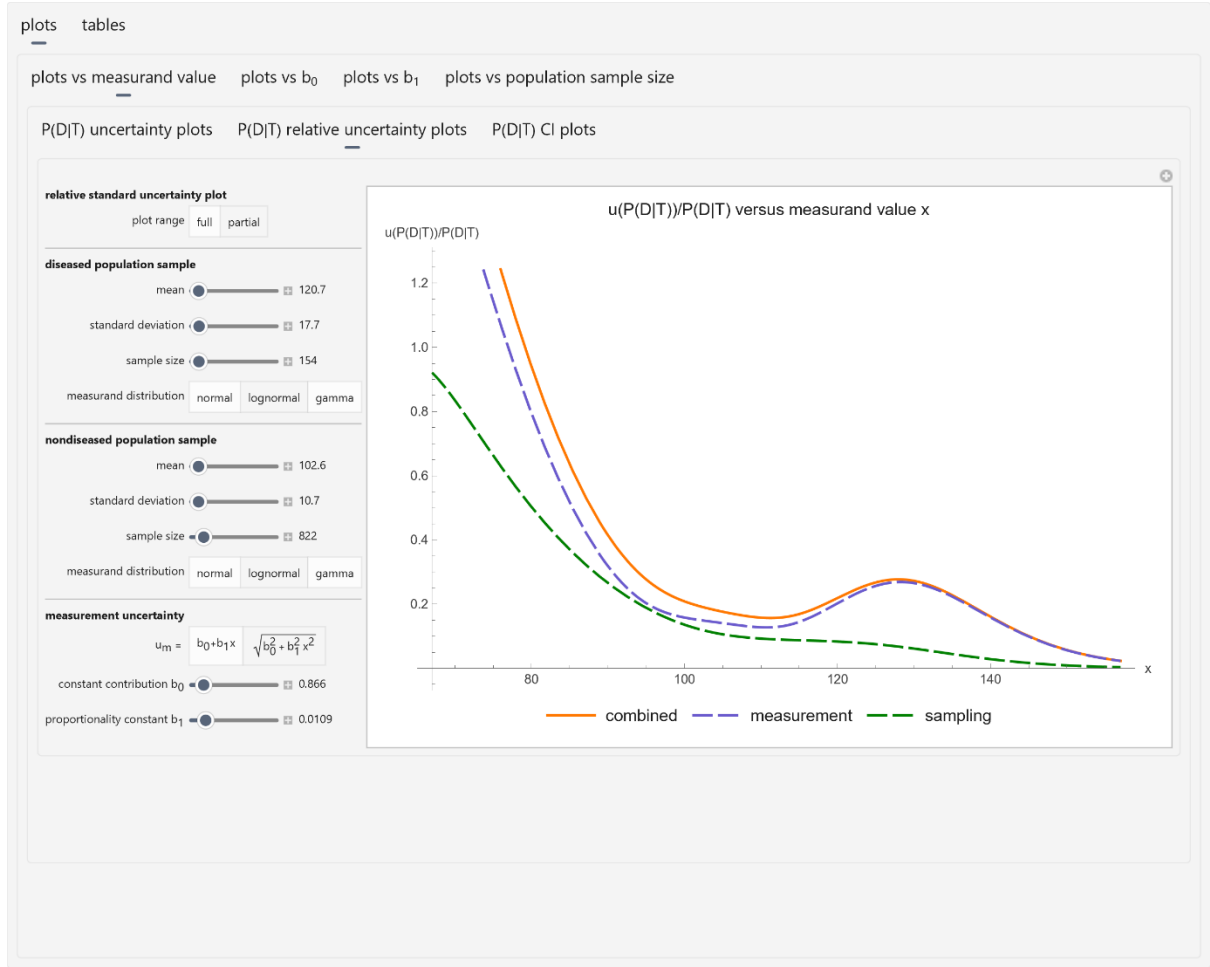


Figure 2. A screenshot of the program *Bayesian Diagnostic Uncertainty*.

3. Illustrative Case Study

To demonstrate the application of the program, fasting plasma glucose (FPG) was used as the diagnostic test measurand for the Bayesian diagnosis of diabetes mellitus (hereafter, ‘diabetes’ refers to diabetes mellitus). The oral glucose tolerance test (OGTT) was the reference diagnostic method. A diagnosis of diabetes was confirmed if the plasma glucose value was equal to or greater than 200 mg/dL, measured two hours after oral administration of 75 g of glucose (ElSayed et al. 2023) during an OGTT (2-h PG). The study population was confined to individuals aged between 70 and 80 years, a decision guided by the well-documented strong correlation between age and the prevalence of diabetes (Sun et al. 2022).

National Health and Nutrition Examination Survey (NHANES) participants data was retrieved for the period from 2005 to 2016 ($n = 60,936$) (National Center for Health Statistics 2005-2016). NHANES is a series of studies designed to evaluate the health and nutritional status of adults and children in the United States.

The inclusion criteria for participants were:

1. Valid fasting plasma glucose (FPG) and oral glucose tolerance test (OGTT) results ($n = 13,836$).
2. A negative response to NHANES question DIQ010 regarding a diabetes diagnosis (National Center for Health Statistics 2005-2016) ($n = 13,465$).
3. Age 70–80 years ($n = 976$).

Participants with a 2-h PG measurement ≥ 200 mg/dL were considered diabetic ($n = 154$).

The prior probability for diabetes was estimated as follows:

$$r = \frac{154}{976} = 0.158$$

The statistics of the FPG datasets are presented in Table 1 (Hereafter, FPG and its uncertainty are expressed in mg/dL).

Table 1. Descriptive statistics of the fasting plasma glucose datasets.

	Diabetic Patients	Nondiabetic Patients
n	154	822
Mean	120.7	102.6
Median	117.0	102.0
Standard Deviation	19.1	10.9
Skewness	1.448	0.523
Kurtosis	6.354	3.445

Lognormal distributions were estimated to model FPG measurements in diabetic and nondiabetic participants using the maximum likelihood estimation method (Myung 2003). The respective distributions, parametrized for their means μ_D and $\mu_{\bar{D}}$, and standard deviations σ_D and $\sigma_{\bar{D}}$, were the following:

$$L_D = \text{Lognormal}(\mu_D, \sigma_D) = \text{Lognormal}(120.671, 17.720)$$

$$L_{\bar{D}} = \text{Lognormal}(\mu_{\bar{D}}, \sigma_{\bar{D}}) = \text{Lognormal}(102.642, 10.653)$$

NHANES quality control data of the FPG measurements was retrieved for the same period (2005–2016). A total of 1,350 QC samples were analysed. The weighted nonlinear least squares analysis (Nielsen 2007) yielded the following function relating the standard measurement uncertainty $u_m(x)$ to the measurement value x :

$$u_m(x) = \sqrt{b_0^2 + b_1^2 x^2} = \sqrt{0.7501 + 0.00012x^2}$$

where $b_0 = 0.866$ and $b_1 = 0.109$.

The means of the standard measurement uncertainty of the FPG of the included diabetic and nondiabetic participants were estimated as follows:

$$\hat{u}_D \approx 1.586 \text{ mg/dL}$$

$$\hat{u}_{\bar{D}} \approx 1.028 \text{ mg/dL}$$

Consequently, the distributions of the measurands, assuming negligible uncertainty, were estimated as:

$$l_D \approx \text{Lognormal}\left(\mu_D, \sqrt{\sigma_D^2 - \hat{u}_D^2}\right) \approx \text{Lognormal}(120.671, 17.720)$$

$$l_{\bar{D}} \approx \text{Lognormal}\left(\mu_{\bar{D}}, \sqrt{\sigma_{\bar{D}}^2 - \hat{u}_{\bar{D}}^2}\right) \approx \text{Lognormal}(102.642, 10.653)$$

Table 2 displays the descriptive statistics of the estimated lognormal distributions of the diabetic and nondiabetic populations, including the p -values of the Cramér–von Mises goodness-of-fit test (Darling 1957).

Table 2. Descriptive statistics of the estimated lognormal distributions of the diabetic and nondiabetic populations.

	Diabetic Participants		Nondiabetic Participants	
Estimated Distribution	L_D	l_D	$L_{\bar{D}}$	$l_{\bar{D}}$
Mean Uncertainty	1.586	0	1.028	0
Mean	120.7	120.7	102.6	102.6
Median	119.4	119.4	102.1	102.1
Standard Deviation	17.8	17.7	10.9	10.7
Skewness	0.446	0.444	0.315	0.312
Kurtosis	3.355	3.352	3.177	3.174
p -value (Cramér–von Mises test)	0.294	0.295	0.281	0.299

Figures 3 and 4 show the estimated PDFs of FPG in the diabetic and nondiabetic populations, assuming a lognormal distribution and negligible measurement uncertainty, and the histograms of the respective NHANES datasets.

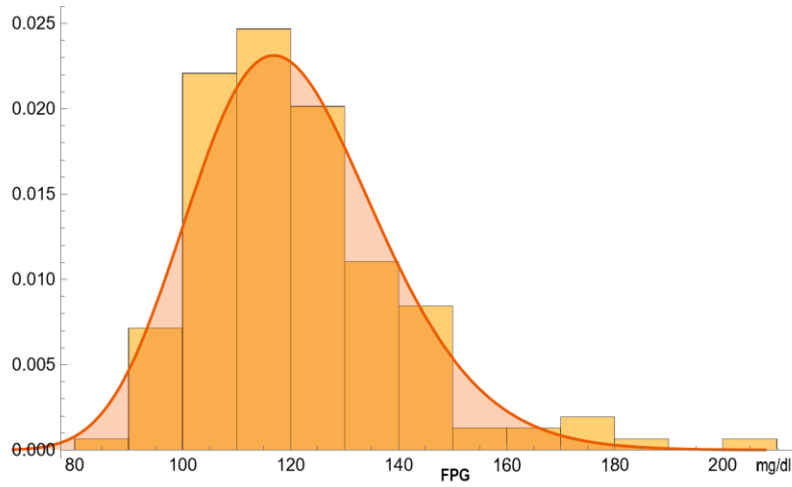


Figure 3. The estimated PDF of the FPG (mg/dL) in diabetic participants, assuming a lognormal distribution and negligible measurement uncertainty, and the histogram of the respective NHANES dataset, with the distribution parameters in Table 2.

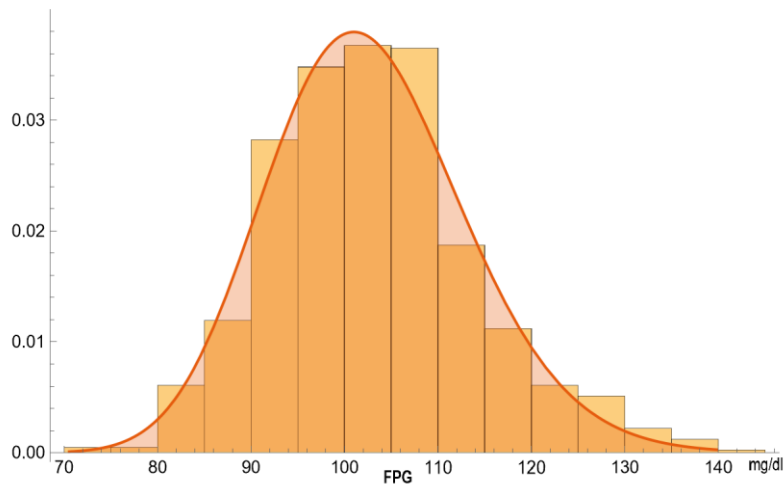


Figure 4. The estimated PDF of the FPG (mg/dL) in nondiabetic participants, assuming a lognormal distribution and negligible measurement uncertainty, and the histogram of the respective NHANES dataset, with the parameters of the distribution in Table 2.

Likelihoods and posterior probabilities were estimated accordingly.

4. Results

Using the settings of Table 3, the program generated the plots of Figures 5–16 and the tables of Figures 17–19.

Table 3. The settings of the program *Bayesian Diagnostic Uncertainty* for Figures 5–19.

Settings	Figures 5 and 6	Figure 7	Figures 8 and 9	Figure 10	Figures 11 and 12	Figure 13	Figures 14 and 15	Figure 16	Figures 17 and 18	Figure 19
p	-	0.95	-	0.95	-	0.95	-	0.95	-	0.95
x	31.0–192.0	31.0–192.0	126.0	126.0	126.0	126.0	126.0	126.0	126.0	126.0
μ_D	120.7	120.7	120.7	120.7	120.7	120.7	120.7	120.7	120.7	120.7
σ_D	17.7	17.7	17.7	17.7	17.7	17.7	17.7	17.7	17.7	17.7
n_D	154	154	154	154	154	154	-	-	154	154
$\mu_{\bar{D}}$	102.7	102.7	102.7	102.7	102.7	102.7	102.7	102.7	102.7	102.7
$\sigma_{\bar{D}}$	10.7	10.7	10.7	10.7	10.7	10.7	10.7	10.7	10.7	10.7
$n_{\bar{D}}$	822	822	822	822	822	822	-	-	822	822
n	-	-	-	-	-	-	65–5000	65–5000	-	-
r	-	-	-	-	-	-	0.158	0.158	-	-
b_0	0.866	0.866	0.0–1.605	0.0–1.605	0.866	0.866	0.866	0.866	0.866	0.866
b_1	0.0109	0.0109	0.0109	0.0109	0.0–0.1	0.0–0.1	0.0109	0.0109	0.0109	0.0109
n_U	-	1350	-	1350	-	1350	-	1350	-	1350
l_D	lognormal	lognormal	lognormal	lognormal	lognormal	lognormal	lognormal	lognormal	normal gamma	normal gamma
$l_{\bar{D}}$	lognormal	lognormal	lognormal	lognormal	lognormal	lognormal	lognormal	lognormal	normal gamma	normal gamma

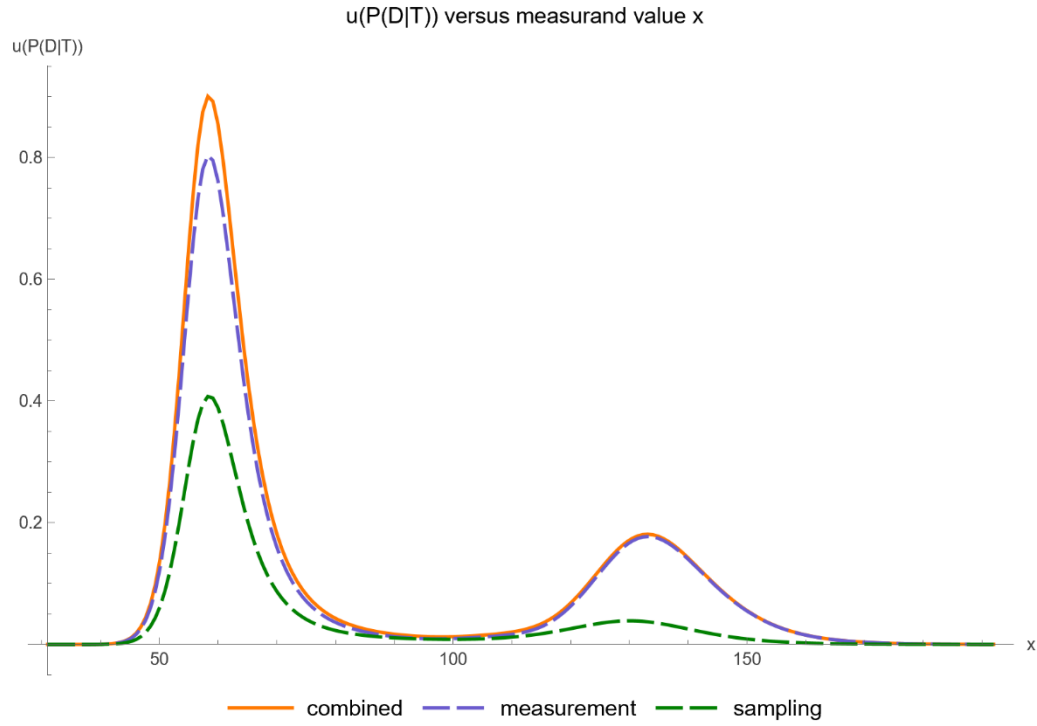


Figure 5. Standard sampling, measurement, and combined uncertainty of the posterior probability for diabetes versus FPG curve plot, with the program's settings in Table 3.

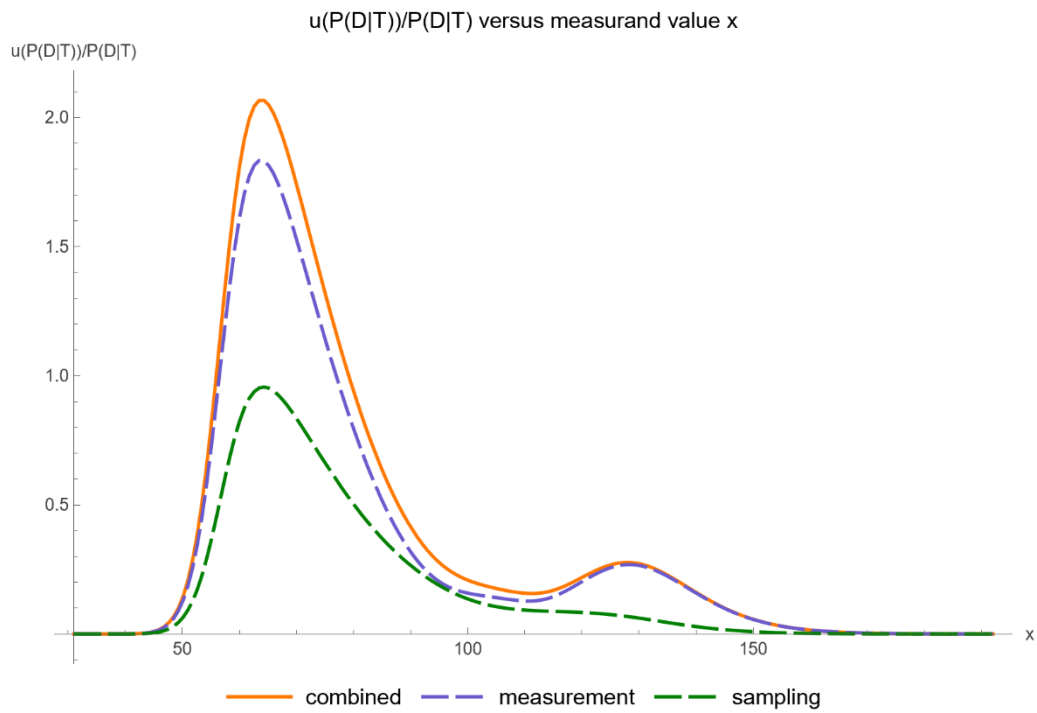


Figure 6. Relative standard sampling, measurement, and combined uncertainty of the posterior probability for diabetes versus FPG curve plot, with the program's settings in Table 3.

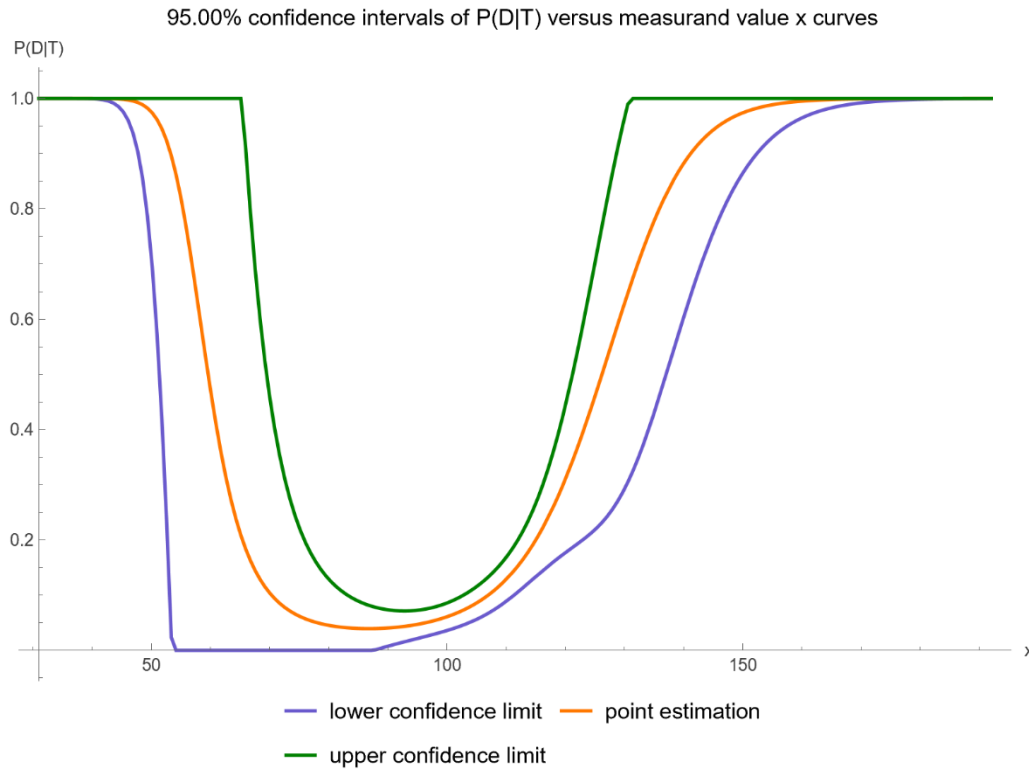


Figure 7. Confidence intervals of the posterior probability for diabetes versus FPG curves plot, with the program's settings in Table 3.

Figure 5 shows the plots of the standard sampling, measurement, and combined uncertainty of posterior probability for diabetes versus FPG, while Figure 6 shows the respective plots of the relative standard uncertainty.

Figure 7 shows the plots of the confidence intervals of posterior probability for diabetes versus FPG for a confidence level $p = 0.95$.

Assessing the combined standard uncertainty of the posterior probability for diabetes, we note the following:

1. It is substantially affected by the measurement uncertainty of FPG.
2. Two local maxima are observed, corresponding to the regions near the steepest segments of the posterior probability curve, which exhibits an approximately double sigmoidal configuration. These maxima are quantitatively defined as follows:
 - 2.1. At an FPG value of 58.5 mg/dL, the posterior probability for disease is equal to 0.581, while the combined standard uncertainty is equal to 0.898.
 - 2.2. At an FPG value of 133.1 mg/dL, the posterior probability for disease is equal to 0.726, while the combined standard uncertainty is equal to 0.181.

This local maxima pattern indicates heightened uncertainty in the regions where the posterior probability curve demonstrates its most pronounced inflections. The confidence intervals are affected accordingly.

Assessing the relative combined standard uncertainty of the posterior probability for diabetes, we note that two local maxima are observed as well, quantitatively defined as follows:

1. At an FPG value of 63.9 mg/dL, the posterior probability for disease is equal to 0.255, while the relative combined standard uncertainty is equal to 2.068.
2. At an FPG value of 128.0 mg/dL, the posterior probability for disease is equal to 0.562, while the relative combined standard uncertainty equals 0.277.

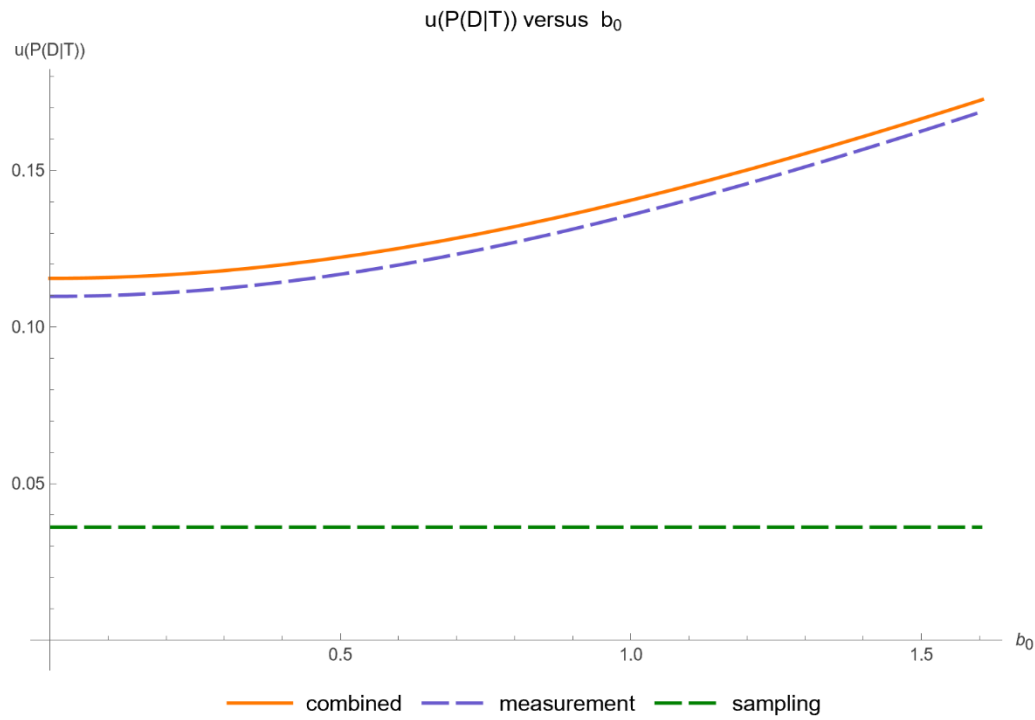


Figure 8. Standard sampling, measurement, and combined uncertainty of the posterior probability for diabetes versus measurement uncertainty constant contribution b_0 curve plot, with the program settings in Table 3.

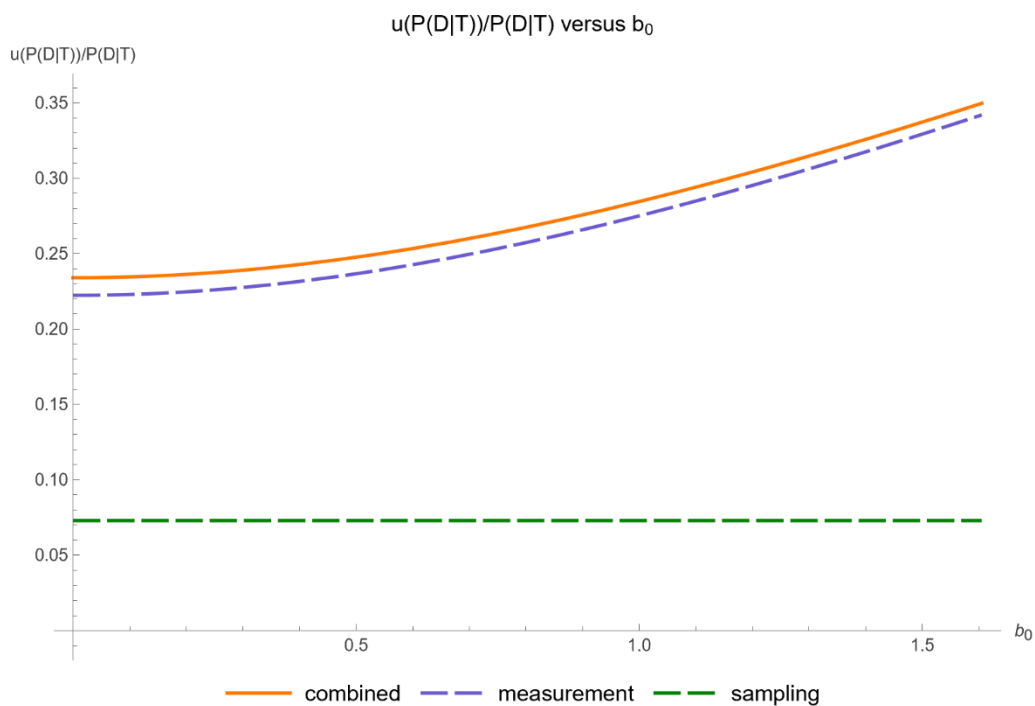


Figure 9. Relative standard sampling, measurement, and combined uncertainty of the posterior probability for diabetes versus measurement uncertainty constant contribution b_0 curve plot, with the program settings in Table 3.

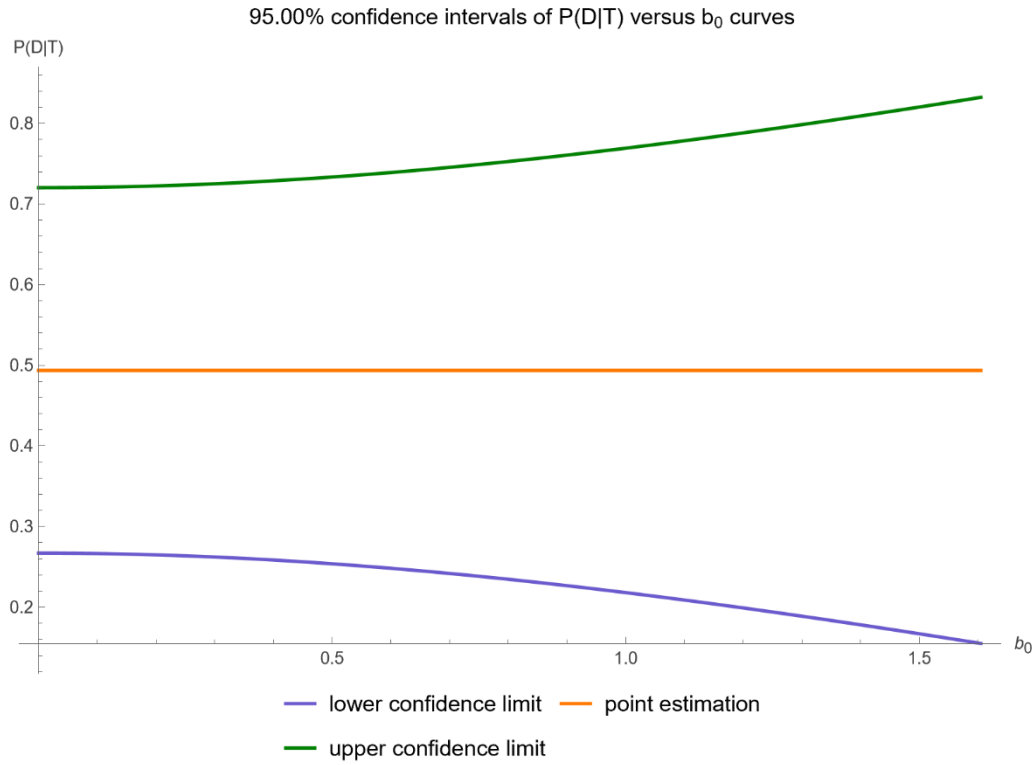


Figure 10. Confidence intervals of the posterior probability for diabetes versus measurement uncertainty constant contribution b_0 curve plot, with the program settings in Table 3.

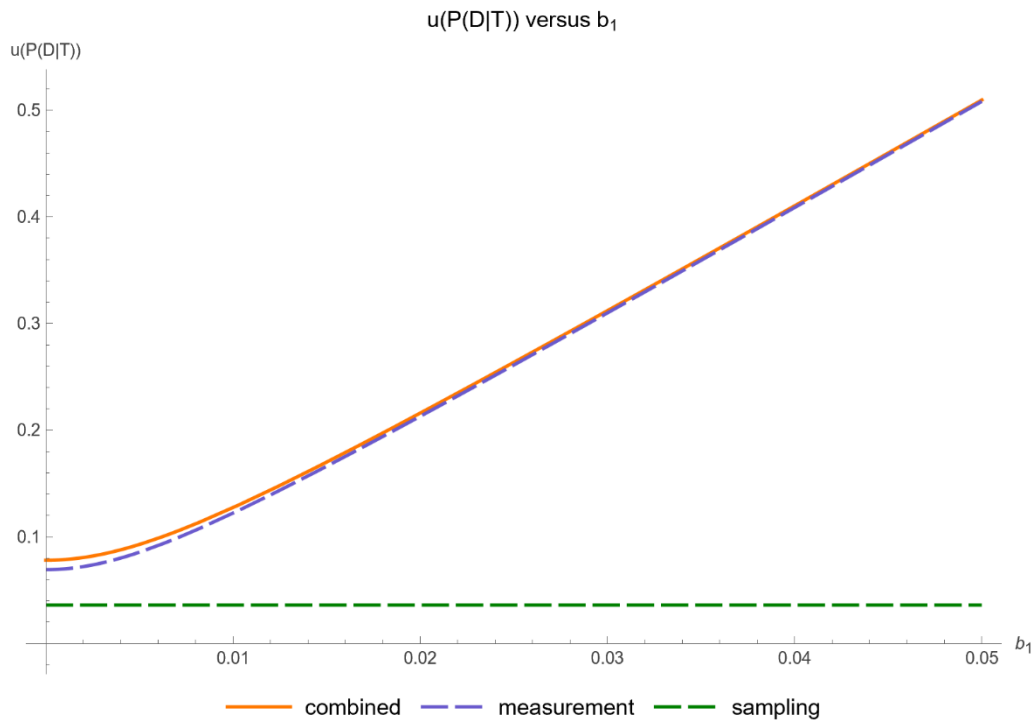


Figure 11. Standard sampling, measurement, and combined uncertainty of the posterior probability for diabetes versus measurement uncertainty proportionality constant b_1 curve plot, with the program settings in Table 3.

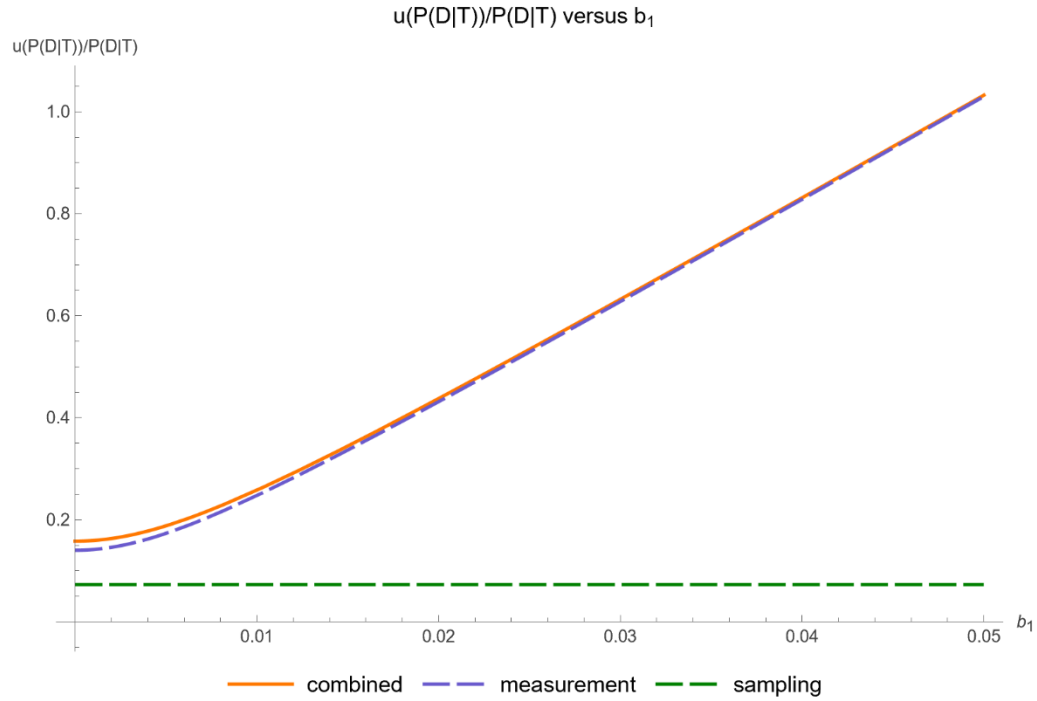


Figure 12. Relative standard sampling, measurement, and combined uncertainty of the posterior probability for diabetes versus measurement uncertainty proportionality constant b_1 curve plot, with the program settings in Table 3.

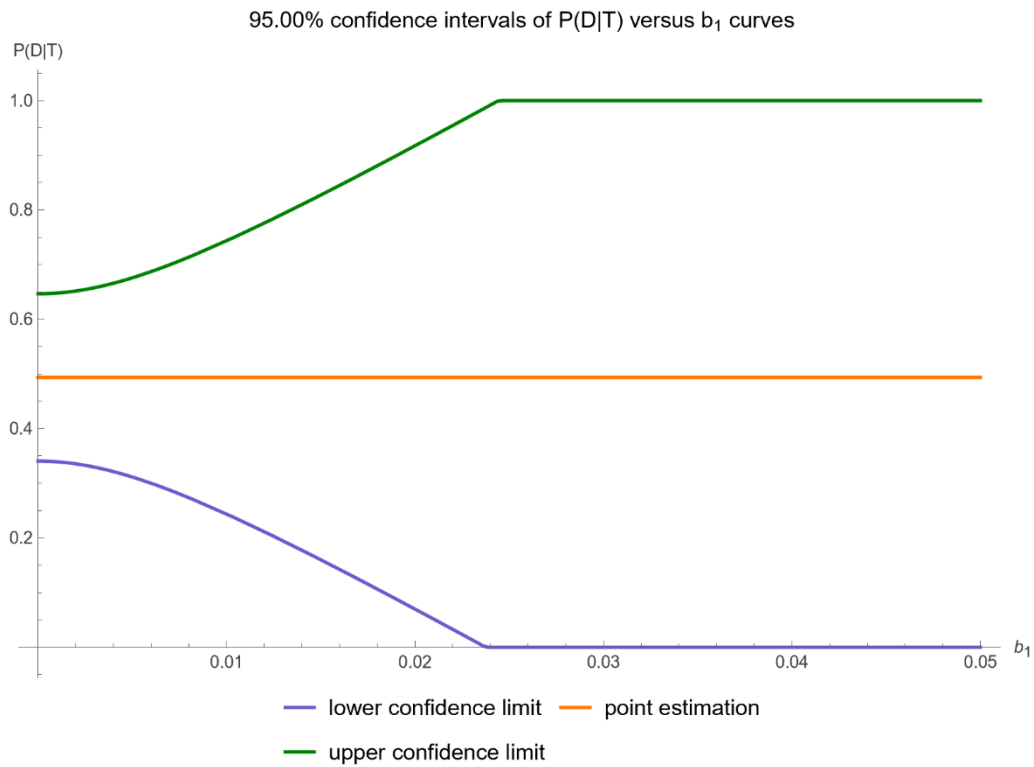


Figure 13. Confidence intervals of the posterior probability for diabetes versus measurement uncertainty proportionality constant b_1 curve plot, with the program settings in Table 3.

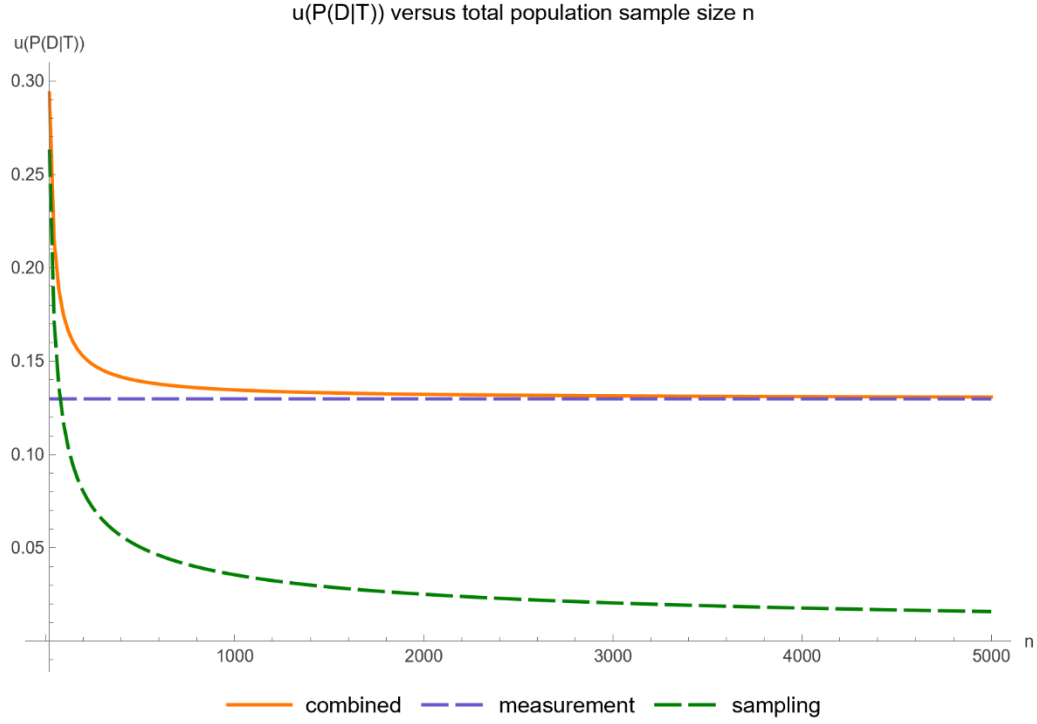


Figure 14. Standard sampling, measurement, and combined uncertainty of the posterior probability for diabetes versus total population sample size n curve plot, with the program's settings in Table 3.

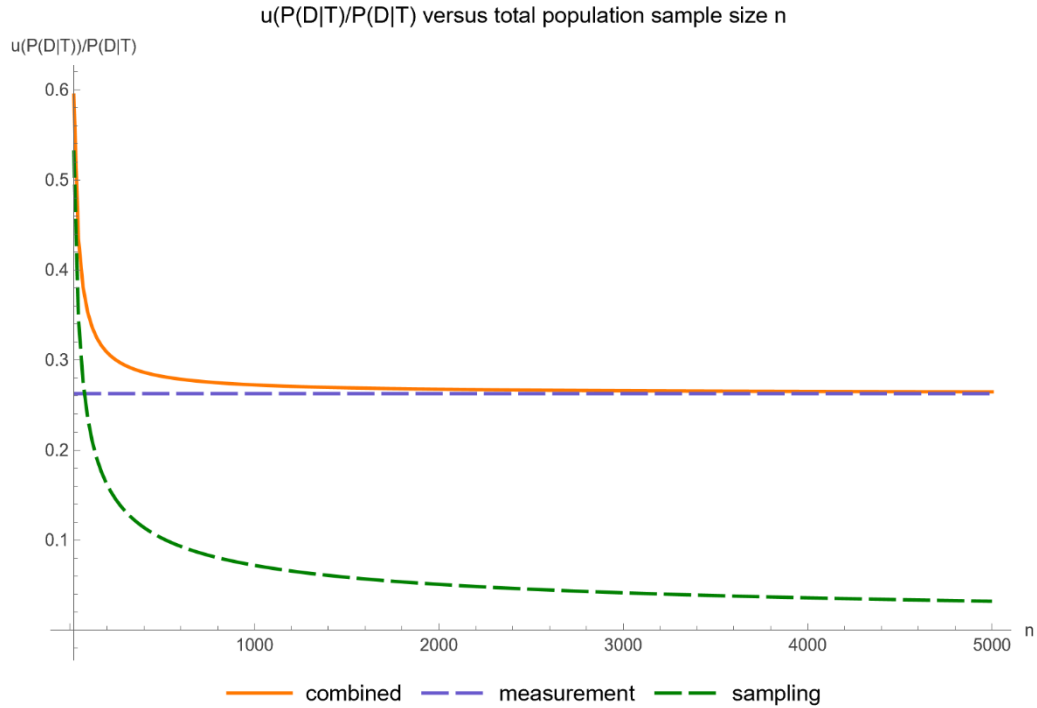


Figure 15. Relative standard sampling, measurement, and combined uncertainty of the posterior probability for diabetes versus total population sample size n curve plot, with the program's settings in Table 3.

Figure 8 shows the plots of the standard sampling, measurement, and combined uncertainty of posterior probability for diabetes versus the constant contribution b_0 of measurement uncertainty of FPG, while Figure 9 shows the respective plots of the relative standard uncertainty.

Figure 10 shows the plots of the confidence intervals of posterior probability for diabetes versus the constant contribution b_0 of measurement uncertainty of FPG, for a confidence level $p = 0.95$.

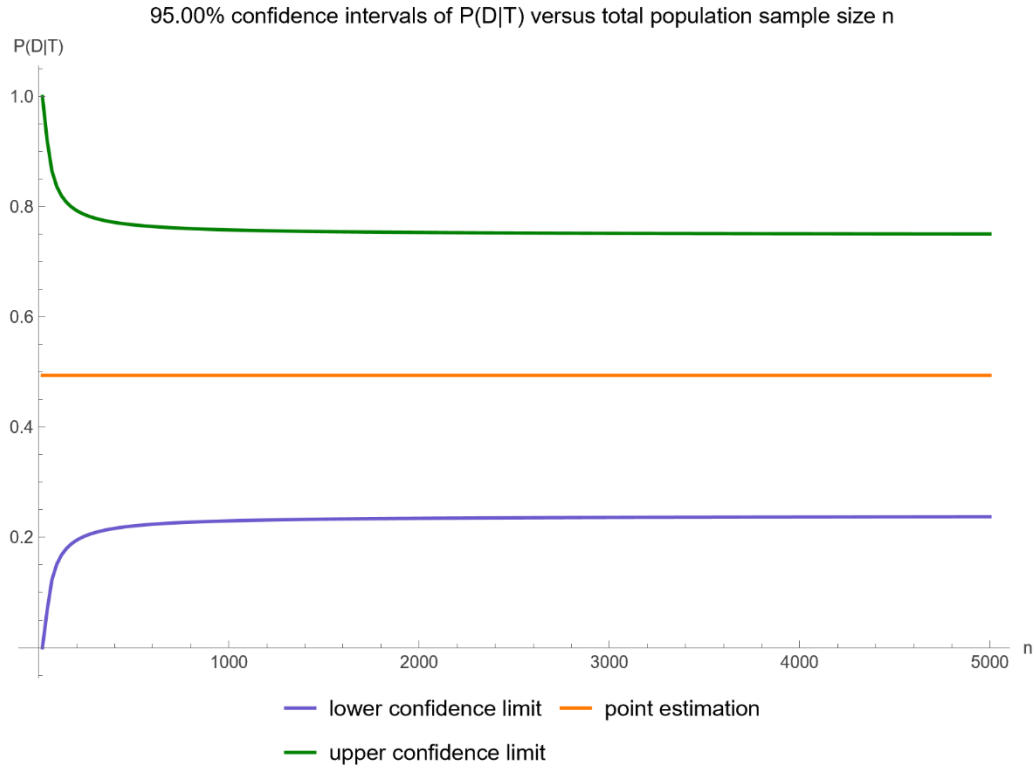


Figure 16. Confidence intervals of the posterior probability for diabetes versus total population sample size n curves plot, with the program's settings in Table 3.

standard uncertainty of posterior probability for disease					
measurand distribution		probability	standard uncertainty		
diseased	nondiseased		combined	measurement	sampling
normal	normal	0.542	0.171	0.166	0.040
	lognormal	0.509	0.134	0.129	0.035
	gamma	0.516	0.145	0.141	0.036
lognormal	normal	0.527	0.135	0.130	0.036
	lognormal	0.494	0.135	0.130	0.036
	gamma	0.501	0.146	0.141	0.037
gamma	normal	0.531	0.172	0.167	0.040
	lognormal	0.498	0.135	0.130	0.036
	gamma	0.505	0.146	0.141	0.037

Figure 17. Standard sampling, measurement, and combined uncertainty of the posterior probability for diabetes, with the program's settings in Table 3.

relative standard uncertainty of posterior probability for disease					
measurand distribution		probability	relative standard uncertainty		
diseased	nondiseased		combined	measurement	sampling
normal	normal	0.542	0.315	0.306	0.073
	lognormal	0.509	0.264	0.254	0.069
	gamma	0.516	0.282	0.273	0.071
lognormal	normal	0.527	0.256	0.246	0.068
	lognormal	0.494	0.273	0.263	0.073
	gamma	0.501	0.291	0.282	0.075
gamma	normal	0.531	0.323	0.314	0.076
	lognormal	0.498	0.270	0.261	0.072
	gamma	0.505	0.289	154.000	0.073

Figure 18. Relative standard sampling, measurement, and combined uncertainty of the posterior probability for diabetes, with the program's settings in Table 3.

posterior probability for disease: 95.00% confidence intervals				
diseased	nondiseased	point estimation	lower confidence limit	upper confidence limit
normal	normal	0.542	0.207	0.877
	lognormal	0.509	0.246	0.772
	gamma	0.516	0.231	0.801
lognormal	normal	0.527	0.263	0.791
	lognormal	0.494	0.229	0.758
	gamma	0.501	0.215	0.787
gamma	normal	0.531	0.195	0.868
	lognormal	0.498	0.234	0.762
	gamma	0.505	0.219	0.791

Figure 19. Confidence intervals of the posterior probability for diabetes, with the program's settings in Table 3.

Figure 11 shows the plots of the standard sampling, measurement, and combined uncertainty of posterior probability for diabetes versus the proportionality constant b_1 of measurement uncertainty of FPG, while Figure 12 shows the respective plots of the relative standard uncertainty.

Figure 13 shows the plots of the confidence intervals of posterior probability for diabetes versus the proportionality constant b_1 of measurement uncertainty of FPG for a confidence level $p = 0.95$.

Figure 14 shows the plots of the standard sampling, measurement, and combined uncertainty of posterior probability for diabetes versus the total population size n , while Figure 15 shows the respective plots of the relative standard uncertainty.

Figure 16 shows the plots of the confidence intervals of posterior probability for diabetes versus the total population size n , for a confidence level $p = 0.95$.

As anticipated, the impact of sampling uncertainty decreases markedly as the size of the population sample increases.

Figure 17 shows a table of the standard sampling, measurement, and combined standard uncertainty of posterior probability for diabetes for FPG value equal to 126 mg/dL, while Figure 18 shows a table of the respective values of relative standard uncertainty.

Figure 19 shows the confidence intervals of posterior probability for diabetes for FPG value equal to 126 mg/dL and confidence level $p = 0.95$.

The tables distinctly demonstrate the considerable magnitude of uncertainty and relative uncertainty associated with the posterior probability for diabetes at an FPG level of 126 mg/dL, the established threshold for the diagnosis of diabetes. Furthermore, the posterior probabilities delineated in

the tables suggest a limited concordance between the classification criteria of diabetes derived from the OGTT and FPG tests (ElSayed et al. 2023), as found previously in the existing literature (Tucker 2020).

5. Discussion

5.1. Reevaluation of Traditional Diagnostic Methods

Traditional diagnostic methods rely on predetermined thresholds; however, this often fails to consider the complexities of disease pathology. While this has been historically effective, it may lack the ability to offer a holistic approach in today's patient-centered medicine, where personalized care is paramount (Obermeyer and Emanuel 2016). The evolving nature of diseases and shifts in patient demographics increase the complexity of the diagnostic process, pushing the boundaries of conventional methodologies. In this challenging context, Bayesian inference emerges as an alternative approach, offering probabilistic evaluations that can adapt to individual patient profiles (Choi, Johnson, and Thurmond 2006; Chatzimichail and Hatjimihail 2023).

Nevertheless, estimating the uncertainty of posterior probabilities within Bayesian inference remains a pivotal challenge (Srinivasan, Westover, and Bianchi 2012). This issue is critically important in the context of diagnostic and screening tests for life-threatening conditions or those associated with considerable morbidity risk. It underscores the need for well-informed clinical judgments and comprehensive uncertainty evaluation in medical decision-making. Key examples include:

1. Cardiac troponin for diagnosing myocardial injury and infarction (Wereski et al. 2021).
2. Natriuretic peptides for the diagnosis of heart failure (Roberts et al. 2015).
3. D-dimer for diagnosing thromboembolic events (Freund et al. 2021).
4. FPG, OGTT, and glycated hemoglobin (HbA1c) for diagnosing diabetes (ElSayed et al. 2023).
5. OGTT for the diagnosis of gestational diabetes (Rani and Begum 2016).
6. Thyroid stimulating hormone (TSH), free serum triiodothyronine (T_3), and free serum thyroxine (T_4) for diagnosing thyroid dysfunction (Reyes Domingo, Avey, and Doull 2019).
7. Protein-to-creatinine ratio for the diagnosis of preeclampsia (Rodriguez-Thompson and Lieberman 2001).
8. Creatinine or cystatin C derived glomerular filtration rate (GFR), and albuminuria for diagnosing chronic kidney disease (Moynihan, Glasscock, and Doust 2013).

The ability to quantify this uncertainty is not only an academic concern but also a practical necessity in improving diagnosis and patient outcomes.

Our software explores the sampling, measurement, and combined uncertainty of Bayesian posterior probabilities to address this. This exploration is vital for enhancing clinical decision-making and plays a significant role in the fields of quality and risk management in laboratory medicine (Haeckel et al. 2016). Additionally, it may contribute to the design and implementation of test accuracy studies (J. Andre Knottnerus and Buntinx 2011; Hajian-Tilaki 2014). As mentioned in the Introduction section, despite the extensive body of research on Bayesian diagnosis and uncertainty as separate entities, the intersection of these two areas remains relatively unexplored (Baron 1994; Ashby and Smith 2000).

The illustrative case study, focusing on individuals aged 70 to 80 years, was designed to mitigate age-related variations in disease prevalence. This focus exemplifies the considerations required in modern diagnostics, where factors such as age, genetics, and lifestyle choices should be accounted for in the diagnostic equation.

Through its analysis of sampling, measurement, and combined uncertainty (as illustrated in Figures 5, 8, 11, 14, and 17), relative uncertainty (Figures 6, 9, 12, 15, and 18), and the corresponding confidence limits (Figures 7, 10, 13, 16, and 19), the software demonstrates its versatility in addressing these diagnostic challenges. Although the software's calculations are highly sophisticated, its user-friendly interface renders it an effective tool for medical researchers and professionals.

The case study from Section 4 highlights the substantial impact of combined uncertainty on the diagnostic process. This finding emphasizes the predominant role of measurement uncertainty and thus stresses the demanding path toward enhancing diagnostic accuracy. By improving the analytical methods of screening and diagnostic tests, the medical community could achieve more accurate diagnoses, leading to more effective and tailored patient care.

Future research should focus on improving the estimations of the uncertainty of posterior probabilities under a diverse array of clinically relevant parameter settings. To transition from research into practical application, it is necessary to focus on clinical decision analysis, studies on cost-effectiveness, and research on quality of care, including implementation studies (J. Andre Knottnerus and Buntinx 2011). Such efforts are necessary to address the complex issues in diagnostic medicine and find new and effective approaches to tackle ongoing challenges.

5.2. Limitations and Future Research Directions

This program's limitations, which provide paths for further research, include:

1. Underlying assumptions:
 - 1.1. The existence of "gold standards" in diagnostics. If a "gold standard" does not exist, there are alternative approaches for classification (J. A. Knottnerus and Dinant 1997; Pfeiffer and Castle 2005; van Smeden et al. 2014).
 - 1.2. The hypothesis of parametric distribution of measurements or their transformations. However, existing literature underlines the robustness of nonparametric techniques in capturing complex data distributions (Wasserman 2006).
 - 1.3. The generally accepted bimodality of the measurands, although unimodal distributions could be considered (J. M. G. Wilson and Jungner 1968; Petersen and Horder 1992).

If these assumptions are not valid, the program may underestimate the standard uncertainty of the posterior probability for disease.

2. Using first-order Taylor series approximations in uncertainty propagation calculations, where higher-order approximations may provide more accurate estimations (Joint Committee for Guides in Metrology 2011).
3. The approximation of the uncertainty of the prior probability for disease using the Agresti-Coull-adjusted Wald interval, despite more accurate methods being available (Pires and Amado 2008).
4. The approximations of the sampling uncertainties for both the sample means and standard deviations, which can be improved for smaller samples or pronounced skewness observed in lognormal and gamma distributions (Schmoyer et al. 1996; Bhaumik, Kapur, and Gibbons 2009).
5. The use of confidence intervals derived from the t -distribution despite the high relative uncertainty (Williams 2020). Though not typical in a Bayesian context, this can be employed instead of credible intervals as a practical tool under certain circumstances (Gelman et al. 2013; Stephens 2023).

While addressing these limitations would considerably increase computational complexity, they represent key areas for future enhancement (Joint Committee for Guides in Metrology 2008, 2020).

5.3. Case Study Shortcomings

The case study's main limitations include reliance on the OGTT as the reference method for diagnosing diabetes mellitus, despite several factors influencing glucose tolerance (Meneilly and Elliott 1999; Geer and Shen 2009; Van Cauter, Polonsky, and Scheen 1997; Colberg et al. 2010; Salmerón et al. 1997; Surwit et al. 2002; Pandit et al. 1993; Dupuis et al. 2010). Additionally, the lognormal distributions used only approximate the distributions of the FPG measurements from NHANES datasets, highlighting the need for more flexible statistical models.

5.4. Challenges in Bayesian Analysis for Disease Diagnosis

While Bayesian analysis may benefit medical diagnostics, it presents specific challenges. For instance, the substantial uncertainty of the posterior probability for disease revealed in our study could lead to clinical indecision. Additionally, a notable lack of comprehensive statistical research on the distribution of measurands in diseased and nondiseased populations hinders further advancements in Bayesian analysis in this field.

5.5. Implications of Incomplete Data

1. Over-reliance on prior probabilities: Limited empirical data may cause an overdependence on prior probabilities, leading to distorted posterior probabilities and potentially flawed clinical decisions (O'Hagan et al. 2006).
1. Increased uncertainty: Insufficient data amplifies the uncertainty of computed posterior probabilities, which could exacerbate clinical indecision (Berger 1985).
2. Bias risks: Unrepresentative datasets could introduce systemic bias, increasing the uncertainty in Bayesian computations (Gelman et al. 2013).

5.6. Analysis of the Double Sigmoidal Curve in Posterior Probability Estimation and Its Impact on Uncertainty

The posterior probability for disease curve, characterized by a double sigmoidal shape featuring two symmetrical sigmoid functions, presents compelling analytical perspectives in medical diagnostic statistics. This configuration implies that the risk associated with the disease may escalate at the lower and

upper extremes of a given measurand, while a zone of relative safety exists in the intermediate range. Notably, the uncertainty associated with the posterior probability for disease becomes markedly pronounced along the steep segments of the double sigmoidal curve. This heightened uncertainty is attributable to minor variations in the measurand value that can lead to significant alterations in the computed posterior probability.

5.7. Software Comparison

Our software easily generates a wide array of parametric plots and comprehensive tables to analyze posterior probability uncertainty. To the best of our knowledge, no software, including all major general or medical or Bayesian statistical and uncertainty quantification software packages (JASP® ver. 0.20.0, Mathematica® ver. 14.0, Matlab® ver. R2023b, MedCalc® ver. 20.2.1, metRology ver. 2023, NCSS® ver. 24.0.0, NIST Uncertainty Machine ver. 2.0.0, OpenBUGS ver. 3.3.0, R ver. 4.3.1, SAS® ver. 9.5, SPSS® ver. 29, Stan ver. 2.33.0, Stata® ver. 19, and UQLab ver. 2.0.0) provides this range of plots and tables without advanced programming.

6. Conclusions

The program we have developed represents a novel approach to estimating and analyzing the uncertainty of Bayesian posterior probabilities in disease diagnosis. This tool stands out not only for its innovative capabilities in medical diagnostics but also as a significant educational and research asset. Considering the difficulties and complexities outlined, this software offers essential assistance in applying Bayesian methods and dealing with diagnostic uncertainties, enhancing well-informed decision-making.

Looking forward, it seems imperative that future research should focus on improving this method with advanced statistical concepts and empirically validating it with comprehensive test accuracy studies. Such studies are essential to verify the efficacy and reliability of the program in real clinical settings. Additionally, expanding its application across a diverse range of diagnostic modalities is necessary. Doing so could enable the program to address a broader spectrum of diagnostic challenges, enhancing its utility and impact in the medical field.

Our research, undertaken alongside our prior work on the uncertainty of diagnostic accuracy measures (Chatzimichail and Hatjimihail 2021), creates a foundation for understanding uncertainties in diagnostic tests. With this consideration, we would recommend employing our approach in diagnostic accuracy research, aiming at formulating clear guidelines and establishing best practices to effectively integrate such information into clinical practice (J. Andre Kottnerus and Buntinx 2011; Whiting et al. 2013; Salameh et al. 2020; Schlattmann 2023).

Regarding regulatory issues, it is necessary to ensure that the software application adheres to the standards set forth by local regulatory authorities.

The potential of this program extends beyond its practical implications in medical diagnostics. As an educational resource, it could offer significant opportunities for training in medical statistics, particularly in understanding the uncertainty of Bayesian posterior probabilities. Its user-friendly interface, coupled with the depth of its analytical capabilities, makes it an effective learning tool for aspiring and experienced medical professionals.

In conclusion, the development and refinement of the *Bayesian Diagnostic Uncertainty* program are pivotal steps toward navigating the complexities of modern medical diagnostics. Its role in enhancing Bayesian diagnostic methods and its educational benefits highlights its capability as a supporting tool in the ongoing evolution of medical practice and research.

Appendix A

Appendix A.1. Formalisms and Notation

Acronyms

PDF: probability density function
 FPG: fasting plasma glucose
 OGTT: oral glucose tolerance test
 NHANES: National Health and Nutrition Examination Survey

Notation

Parameters

n_D : size of diseased population
 μ_D : mean of diseased population
 σ_D : standard deviation of diseased population
 $n_{\bar{D}}$: size of nondiseased population

$\mu_{\bar{D}}$: mean of nondiseased population
 $\sigma_{\bar{D}}$: standard deviation of nondiseased population
 r : prior probability for disease (prevalence rate)
 $u_s(x)$: standard sampling uncertainty of x
 $u_m(x)$: standard measurement uncertainty of x
 $u_c(x)$: standard combined uncertainty of x
 n_U : number of quality control measurements
 b_0 : constant contribution to measurement uncertainty
 b_1 : measurement uncertainty proportionality constant
 $\hat{u}$: mean standard measurement uncertainty
 p : confidence level
 v_{eff} : effective degrees of freedom

Functions

$P(A)$: probability of the event A
 $P(A|B)$: conditional probability of the event A given the event B
 $L(\theta|z)$: likelihood function
 $F^{-1}(\cdot)$: the inverse function $F(\cdot)$

Bayes' Theorem

For the purposes of our study, Bayes' theorem is formulated as:

$$P(D|T) = \frac{P(T|D)P(D)}{P(T)} = \frac{P(T|D)P(D)}{P(T|D)P(D) + P(T|\bar{D})(1 - P(D))}$$

where

$P(D|T)$ denotes the posterior probability of having a disease D given a test result T .
 $P(T|D)$ denotes the likelihood of obtaining the result T given the presence of the disease D .
 $P(T|\bar{D})$ denotes the likelihood of obtaining the result T given the absence of the disease D .
 $P(D)$ is the prior probability or prevalence r of the disease D .
 $P(T)$ is the overall probability of the result T .

According to Bayes' theorem, the posterior probability for a disease D given a test result $T = x$ and a parameter vector θ is calculated as:

$$P(D|T) = \frac{L_D(\theta|x)r}{L_D(x|\theta)r + L_{\bar{D}}(x|\theta)(1 - r)} = \frac{f_D(x|\theta)r}{f_D(x|\theta)r + f_{\bar{D}}(x|\theta)(1 - r)}$$

where r denotes the prior probability for disease, $L_D(\theta|x)$ and $f_D(x|\theta)$ denote the likelihood function and the PDF of the test measurand in the presence of the disease, respectively, while $L_{\bar{D}}(x|\theta)$ and $f_{\bar{D}}(x|\theta)$ are the respective functions in the absence of the disease.

Appendix A.1.1. Parametric Distributions

The test measurands of the diseased or nondiseased populations are assumed to follow the normal, lognormal, or gamma distribution. The domains of random variables for the respective distributions are defined as follows:

1. The domain of a random variable X following a normal distribution is the set of all real numbers, denoting $-\infty < X < \infty$.
2. The domain of a random variable X following a lognormal distribution is the set of all positive real numbers, denoting $0 < X < \infty$.
3. The domain of a random variable X following a gamma distribution is the set of all positive real numbers, denoting $0 < X < \infty$.

Appendix A.1.2. Calculations of the Posterior Probability for Disease and Its Uncertainty

These calculations are detailed in Supplementary File II (Refer to Supplementary Files).

Appendix A.2. Software Availability and Requirements

Program name: *Bayesian Diagnostic Uncertainty*

Version: 2.1.1

DOI: [10.5281/zenodo.21891843](https://doi.org/10.5281/zenodo.21891843)

Project home page: <https://www.hcsl.com/Tools/BayesianDiagnosticUncertainty/> (accessed on August 11, 2026). Available at: <https://www.hcsl.com/Tools/BayesianDiagnosticUncertainty/BayesianDiagnosticUncertainty.nb> (accessed on August 11, 2026)

Operating systems: Microsoft Windows 10+, Linux 3.15+, Apple macOS 11+

Programming language: Wolfram Language

Other software requirements:

To run the program and read the BayesianDiagnosticUncertaintyCalculations.nb file Wolfram Player® ver. 12.0+ is required, freely available at: <https://www.wolfram.com/player/> (accessed on May 15, 2026) or Wolfram Mathematica® ver. 12.0+

System requirements: Intel® i9™ or equivalent CPU and 32 GB of RAM

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Appendix A.3. A Note about the Program

About the Program Controls

The program features an intuitive tabbed interface designed to streamline user interaction and facilitate effortless navigation across multiple modules and sub-modules.

The user defines the numerical settings with menus or sliders. Sliders can be finely manipulated by holding down the *alt* key or *opt* key while dragging the mouse. They are even more finely manipulated by also holding the *shift* and/or *ctrl* keys.

Dragging with the mouse while pressing the *ctrl*, *alt*, or *opt* keys zooms plots in or out.

Range of input parameters

x : $0.01 - \text{maximum}(\mu_{\bar{D}} - 5\sigma_{\bar{D}}, 0) - \mu_D + 5\sigma_D$

n_D : 2–10,000

μ_D : 0.1–10,000

σ_D : 0.01–1000

$n_{\bar{D}}$: 2–10,000

$\mu_{\bar{D}}$: 0.1–10,000

$\sigma_{\bar{D}}$: 0.01–1000

r : 0.001–0.999

n_U : 20–10,000

b_0 : $0 - \sigma_{\bar{D}}$

b_1 : 0–0.1

p : 0.900–0.999

8. Supplementary Files

8.1 Supplementary File I

BayesianDiagnosticUncertainty.nb: The program as a Wolfram Mathematica Notebook. Available at <https://www.hcsl.com/Tools/BayesianDiagnosticUncertainty/BayesianDiagnosticUncertainty.nb>

8.2 Supplementary File II

BayesianUncertaintyCalculations.nb: The calculations for estimating Bayesian posterior probability for disease and its standard uncertainty in a Wolfram Mathematica Notebook. Available at <https://www.hcsl.com/Supplements/SBDU.zip>

8.3 Supplementary File III

BayesianDiagnosticUncertaintyInterface.pdf: A brief description of the program interface. Available at: <https://www.hcsl.com/Documents/BayesianDiagnosticUncertaintyInterface.pdf>

9. Statements

9.1 Institutional Review Board Statement

Data collection was carried out following the rules of the Declaration of Helsinki. The National Center for Health Statistics Ethics Review Board approved data collection and posting of the data online for public use. National Center for Health Statistics NHANES—NCHS Research Ethics Review Board Approval (Protocols #2005-06 and #2011-17), available online at: <https://www.cdc.gov/nchs/nhanes/about/erb.html> (accessed on May 15, 2026).

9.2 Informed Consent Statement

Written consent was obtained from each subject participating in the survey.

9.3 Data Availability Statement

The data presented in this study are available at <https://www.cdc.gov/nchs/nhanes/default.aspx> (accessed on May 15, 2026).

9.4 Conflicts of Interest

The authors declare no conflict of interest.

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